Lab 2: Descriptive Statistics - Central Tendency & Data Visualization

Sleep, Test Anxiety, and Memory During Exam Week

2026-09-10

Overview & Scenario

A psychology research team is studying student well-being during exam week. They recruited 24 undergraduate students and recorded each participant's self-reported sleep from the prior night, test-anxiety score, performance on a short memory-recall task, and primary coping strategy.

The team's research question is:

> **During exam week, what are the typical levels of sleep, test anxiety, and memory recall among undergraduate students, and which coping strategy is most common?**

This is a **descriptive** question. We will use the mean, median, and mode to describe the sample appropriately. We will also use visualizations to help us understand the distributions of the variables. Finally, we will write a concise APA-style paragraph to summarize the results.

Learning Objectives

By the end of this lab, you should be able to:

1. Import an excel format dataset into SPSS and set up variable names, value labels, and measurement levels.
2. Use SPSS to run descriptive summary statistics (central tendency) and basic data visualizations for both continuous and discrete variables.
3. Explain the effect of extreme scores on measures of central tendency such as the mean and the median.

4. Perform simple transformations on continuous variables and explain the effect of these transformations on measures of central tendency.
5. Write a concise APA-style descriptive-results paragraph.

Part A — Importing Data into SPSS

- Download the Exam Week Well Being Dataset. Note where the file is located (most likely downloads).
- Open the SPSS program on your computer. Select **File > Import Data > CSV Data**. Navigate to the file location and select it.
- Click **Open**. In the pop-up window, ensure that **Read variable names from the first row of data** is checked. Click **OK**. An output pop up window may appear but you can ignore it and proceed to the data window. The dataset should now appear in SPSS.
- In **Variable View**, enter the **value labels** and **measurement levels** below.

Variable	Description / labels	Measure
`Participant_id`	Participant identification number	Nominal
`Sleep_Hours_Prior_Night`	Hours slept the night before the exam	Scale
`Memory_Recall_Score`	Number of words recalled correctly; possible range 0–20	Scale
`Test_Anxiety_Score`	Test-anxiety score; possible range 0–100, higher = more anxiety	Scale
`Primary_coping_strategy`	1 = Exercise; 2 = Social Support; 3 = Avoidance; 4 = Planning; 5= Cognitive Reframing	Nominal

Checkpoint

- Double check that you have entered the value labels and measurement levels correctly.
- Save the data file as `lab_2_yourlastname.sav`.

Part B — Analyzing Continuous Variables

Central Tendency

We want to find the "typical hours of sleep" (a continuous variable) for the sampled students by finding the **Mean**, **Mode**, and **Median** using SPSS. Follow the following steps:

- **Analyze > Descriptive Statistics > Frequencies**.
- Move `Sleep_Hours_Prior_Night` to the Variable(s) box. Check the **Display frequency tables** box.
- Click **Statistics**, and check boxes for **Mean**, **Median**, and **Mode**
- Click **Continue**, then **OK**.

The output should show both the frequency table and the three measures of central tendency for `Sleep_Hours_Prior_Night`.

> Checkpoint
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> 1. Report the three measures of center.
> 2. Which of the three measures is the most reliable and why?

Visualizing Continuous Variables (Sleep Hours)

To visualize the distribution of a continuous variable such as `sleep_hours`, a histogram is one of the most reliable tools. Follow the following steps to create a histogram in SPSS:

- **Analyze > Descriptive Statistics > Frequencies**.
- Move `Sleep_Hours_Prior_Night` to the Variable(s) box.
- Click **Charts**, select **Histograms**, and check **Display normal curve**.
- Click **Continue**, then **OK**. Keep the histogram in your output.

> Checkpoint
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> How would you describe the distribution of sleep hours? Is the histogram roughly symmetric, positively skewed, or negatively skewed? The overlayed normal curve can help you determine this.

Another tool that can be used to visualize the distribution of a continuous variable is a box plot. Follow the following steps to create a box plot for `Sleep_Hours_Prior_Night` in SPSS:

- **Graphs > Chart Builder**.
- Select **Boxplot** from the gallery, then drag the simple box plot icon to the chart preview area.

- Drag `Sleep_Hours_Prior_Night` to the y-axis and click **OK**. Keep the box plot in your output.

> Checkpoint
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> How would you describe the distribution of sleep hours based on the box plot?

Extreme Scores

Presence of extreme scores (suspected outliers) can influence central tendency. Change the sleep hours for the last participant (id = 24) to 22 hours. Re-run the analysis for `Sleep_Hours_Prior_Night` and record the new mean, median, and mode. - How does this change affect these values? In this case, which value is more reliable? - How do the distributions change based on the histogram and box plot?

Simple Transformations on Continuous Variables

Now, imagine the research team revises the anxiety scores by adding 5 points to every participant's score. We can do this in SPSS as follows:

- **Transform > Compute Variable**.
- Under ***Target Variable***, write `anxiety_plus_5`.
- Under Numeric Expression write `Test_Anxiety_Score + 5`.
- Click **OK**.

Look at the data file and you should see a new variable called `anxiety_plus_5`.

Run the Mean and median for for `Test_Anxiety_Score` and `anxiety_plus_5`.

> Checkpoint
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> How did the mean and median anxiety change after adding 5 to each individual score? State the rule for what happens to the mean and median when the same constant is added to every score.

Part C — Analyzing Discrete Variables

In most cases, summary statistics such as the mean and median are not useful for discrete variables measured at the nominal or ordinal level. This is because it doesn't make mathematical sense to add two categories (e.g Male +Female). For these variables, the **mode** can be used

to tell us the category with the highest representation (frequency). A frequency distribution table can be made to identify this.

Frequency Distribution Table for Discrete Variables

A frequency Distribution table is a straight forward tool that can be used to summarize and organize a discrete nominal variable such as `Primary_Coping_Startegy`. To make the table in SPSS, follow the following steps:

[Be sure to clear the previous analyses before running this one. For example, disable Histograms]

- **Analyze > Descriptive Statistics > Frequencies**.
- Move `Primary_Coping_Strategy` to the Variable(s) box.
- Check the **Display frequency tables** box.
- Click **OK**.

Visualizing Discrete Variables

To visualize a discrete variable such as `Primary_Coping_Strategy`, a **bar chart** is one of the most reliable tools. Follow the following steps to create a bar chart in SPSS:

- **Analyze > Descriptive Statistics > Frequencies**.
- Move `Primary_Coping_Strategy` to the Variable(s) box.
- Click **Charts**, select **Bar charts**,

- Uncheck the **Display frequency tables** box.
- Click **OK**.

Keep the bar chart in your output.

> Tip
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> Based on the frequency table and bar chart, which coping strategy was the most common among the sample? Report the mode in words, along with its frequency and percentage.

Part D — Analyzing Multiple Variables

Although analyzing variables separately is useful, the complexity of life often require us to look at multiple variables together. For example, we may want to know if students who sleep more hours have lower test anxiety scores. To do this, we can use a scatterplot to visualize the relationship between two continuous variables.

Here is how to create a scatterplot for sleep hours vs test anxiety scores in SPSS:

- **Graphs > Chart Builder**.
- Select **Scatter/Dot** from the gallery, then drag the simple scatter icon to the chart preview area.
- Drag `Sleep_Hours_Prior_Night` to the x-axis and `Test_Anxiety_Score` to the y-axis. Click **OK**. Keep the scatterplot in your output.

We may also want to compare the mean test anxiety scores by coping strategy. To do this, we can use a bar chart to visualize the mean test anxiety scores for each coping strategy.

- **Graphs > Chart Builder**.
- Select **Bar** from the gallery, then drag the simple bar icon to the chart preview
- Drag `Primary_Coping_Strategy` to the x-axis and `Test_Anxiety_Score` to the y-axis. Click **OK**. By default, SPSS will calculate the mean for each group. Keep the bar chart in your output.

Checkpoint

- Based on the scatterplot, is there a relationship between sleep hours and test anxiety scores?
- Based on the bar chart, which coping strategy is associated with the lowest mean test anxiety score?
- Save your SPSS output file as `Lab_2_Output_YourLastName.spv`.

APA-Style Reporting Tips

APA-style results writing should be accurate, concise, and tied to the research question. Use the statistics you produced in SPSS—not estimates from the graph.

- Report the sample size as N when referring to the whole sample.
- Report the mean as M and the standard deviation as SD.
- Report the median as Mdn when it is the better summary for a skewed distribution.
- Report a categorical mode in words, followed by its frequency (n) and percentage when useful.
- Round consistently, typically to two decimal places unless your instructor directs otherwise.
- Describe the distribution when that explains why you selected the median instead of the mean.

Build your Results Paragraph

Use this template to write a paragraph that answers the research question:

> The sample ($N =$ ***) reported an average of M =*** hours of sleep ($SD =$ ***; Mdn =***). Participants recalled an average of $M =$ _____ words ($SD =$ ***) on the memory task. The modal coping strategy was [strategy]*** ($n =$, ***%). Test-anxiety scores were [roughly symmetric / positively skewed / negatively skewed]; therefore, the [mean / median] was the most appropriate indicator of a typical score (M*** **or** ***Mdn =***). Overall, these results suggest that [one sentence summarizing the sample descriptively without making a causal claim].

Take-Home Lab Exercises

1. Use SPSS to generate the mean, median, and mode for the `Memory_Recall_Score` variable.
2. Create a histogram to visualize the distribution of `Memory_Recall_Score`. Describe the distribution of `Memory_Recall_Score` based on the histogram.
3. Create a box plot to visualize the distribution of `Memory_Recall_Score`. Describe the distribution of this variable based on the box plot. What insights can you draw based on the box plot that would be difficult to see in the histogram?
4. Use SPSS to generate a bar graph comparing the mean memory recall across the five coping strategies. Comment on which coping strategy is associated with the highest mean memory recall score.

Submission checklist

- ☐ A saved SPSS data file with correct value labels and measurement levels.
- ☐ SPSS output containing all the analyses performed.
- ☐ Completed responses for the Questions above. This document should be typed (Times New Roman, Font size 12, double spaced). The document should include screenshots of the relevant software outputs where necessary.